# Introduction to Data Warehouse and Dimensional Modeling

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# Last Week's Hospital Case

- One Excel file (Hospital Waiting Time)
- Performed EDA
- Answered waiting time questions
- Distribution & outlier analysis

Hospital_wait.xlsx
↓
EDA
↓
Business Insight



# The hospital has expanded …

As the hospital grows, business questions become much more complex.

**CEO Questions:**

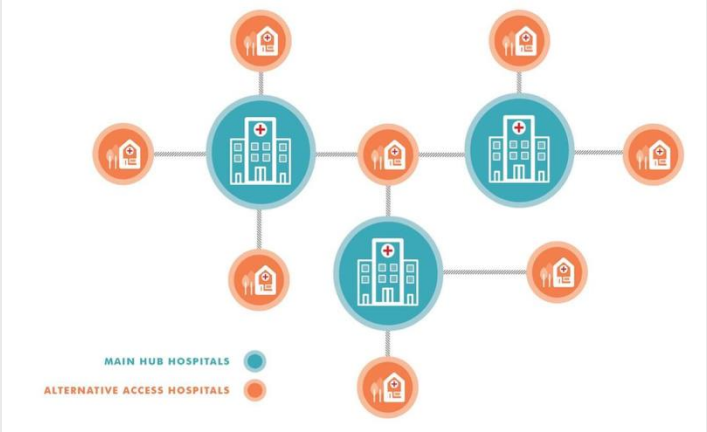
- Revenue by insurance company (Compare revenues)
- Waiting time by branch (Compare branches)
- Doctor performance trend (Analyze trends)
- Patient satisfaction over five years (KPI dashboard)



# Where Are the Data?

Each department has its own database.

- Registration, Appointment
- Laboratory
- Pharmacy
- Billing
- Insurance
- HR
- Patient Survey

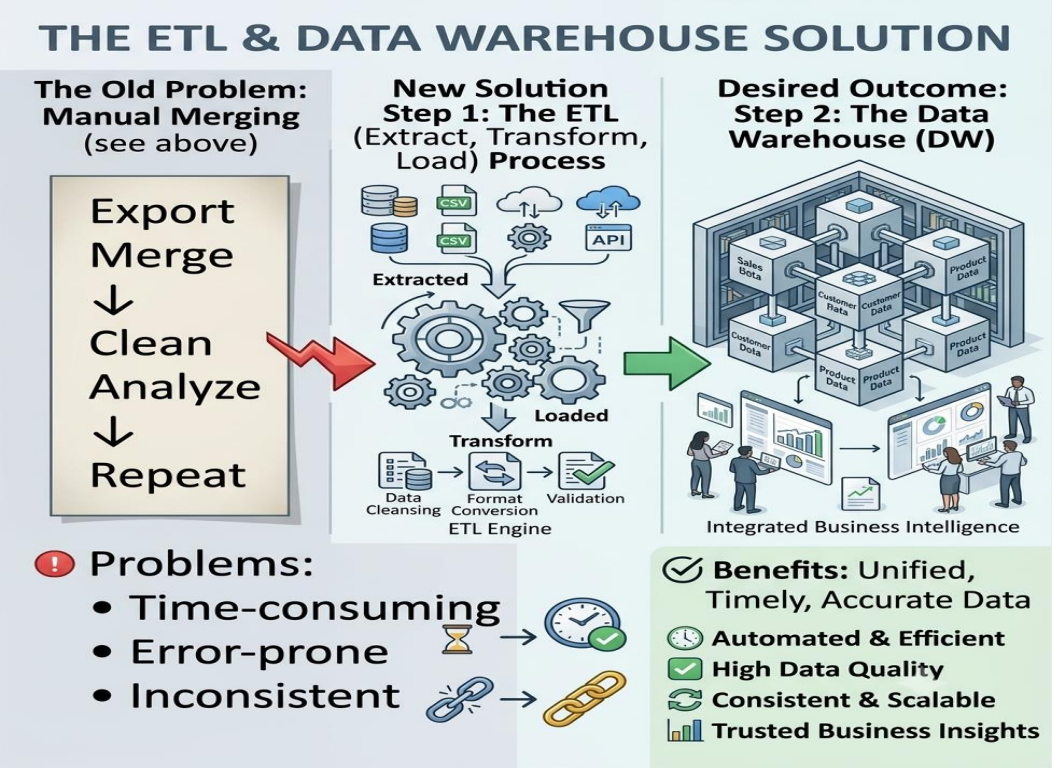



- The problem is NOT Excel, The problem is business complexity.
  - More departments
  - More systems
  - More reports
  - More historical data

# Why Not Just Merge Excel Files?

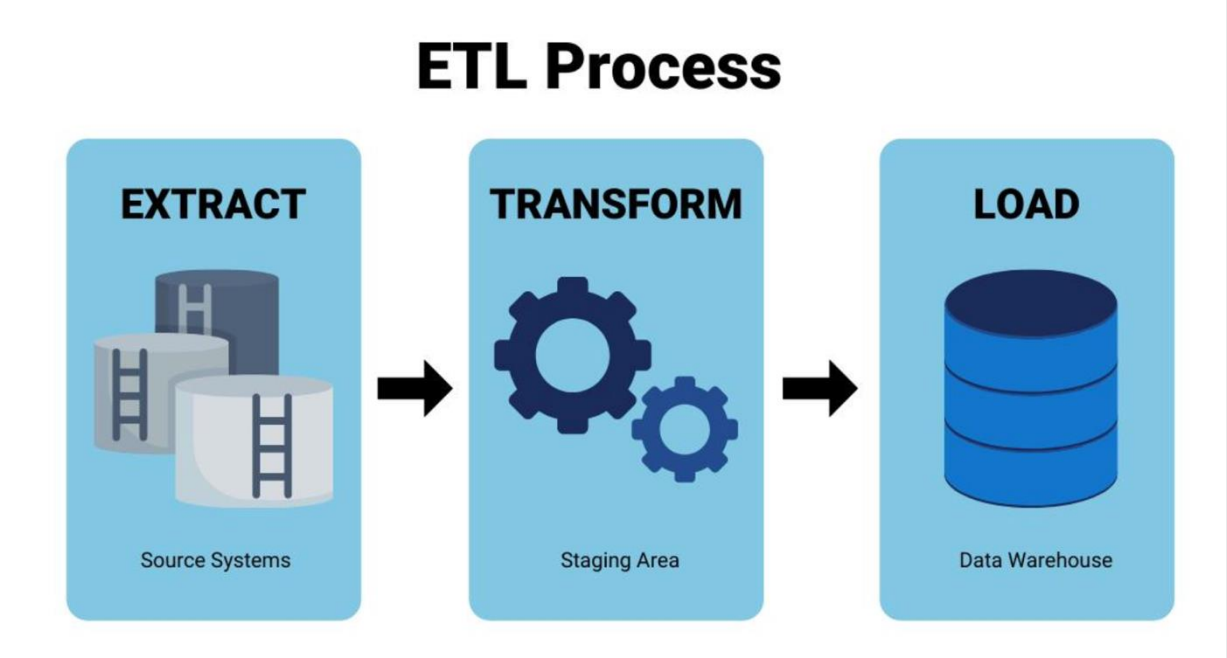
Export
↓
Merge
↓
Clean
↓
Analyze
↓
Repeat

**Problems:**
- Time-consuming
- Error-prone
- Inconsistent



# We Need a Better Solution

Operational Systems
↓
ETL (Data Integration)
↓
Data Warehouse
↓
Dashboards
↓
Decision Making



# Concepts Discussed

- Introduction to Data Warehouse
- Different Information Worlds
- Goals of Data Warehouse
- Components of a Data Warehouse
- Dimensional Modeling Vocabulary

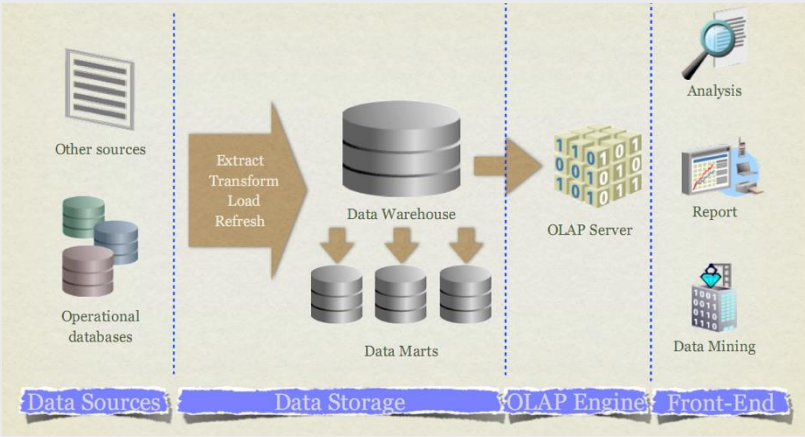
# From EDA to Data Warehouse & Star Schema

- Introduction to **Data Warehouse**:
  - What a Data Warehouse is and why it's important.
- **Multidimensional Design** and **Star Schema**:
  - Explanation of multidimensional design.
  - How Star Schema structures data and its benefits.
- **Benefits of Data Warehouse** for Analytics:
  - Supporting **Descriptive Analytics** & **making data analysis more efficient**.
- **Practical Use Case => Retail Sale**

- **These concepts prepare you for more advanced analytics.**

# What is a data warehouse?

- A **central database** that stores historical business data **from multiple systems**.
  - Combines data from various operational systems across the organization
  - Data is cleaned, organized, and summarized for **Easy Analysis**.
- Designed for **business decision-making**
  - Supports strategic planning and analysis across all departments.
  - Enables users to make informed decisions based on historical trends.
- **Optimized for reporting** and **analysis**
  - Data is structured to make queries fast and efficient.
  - Historical information is preserved for trend analysis and forecasting.



# What is a data warehouse?

- **A centralized repository that:**
  - Aggregates data from multiple sources
  - Creates a single source of truth
  - Stores historical data for analysis
  - Supports analytics and business intelligence
- **Key Benefits:**
  - Centralize data from disparate sources
  - Facilitate smarter decisions using BI
  - Leverage all company data

- **Data warehouses are a part of almost every industry, including e-commerce, transportation, medical, banking and fin-tech, social media, and governments.**

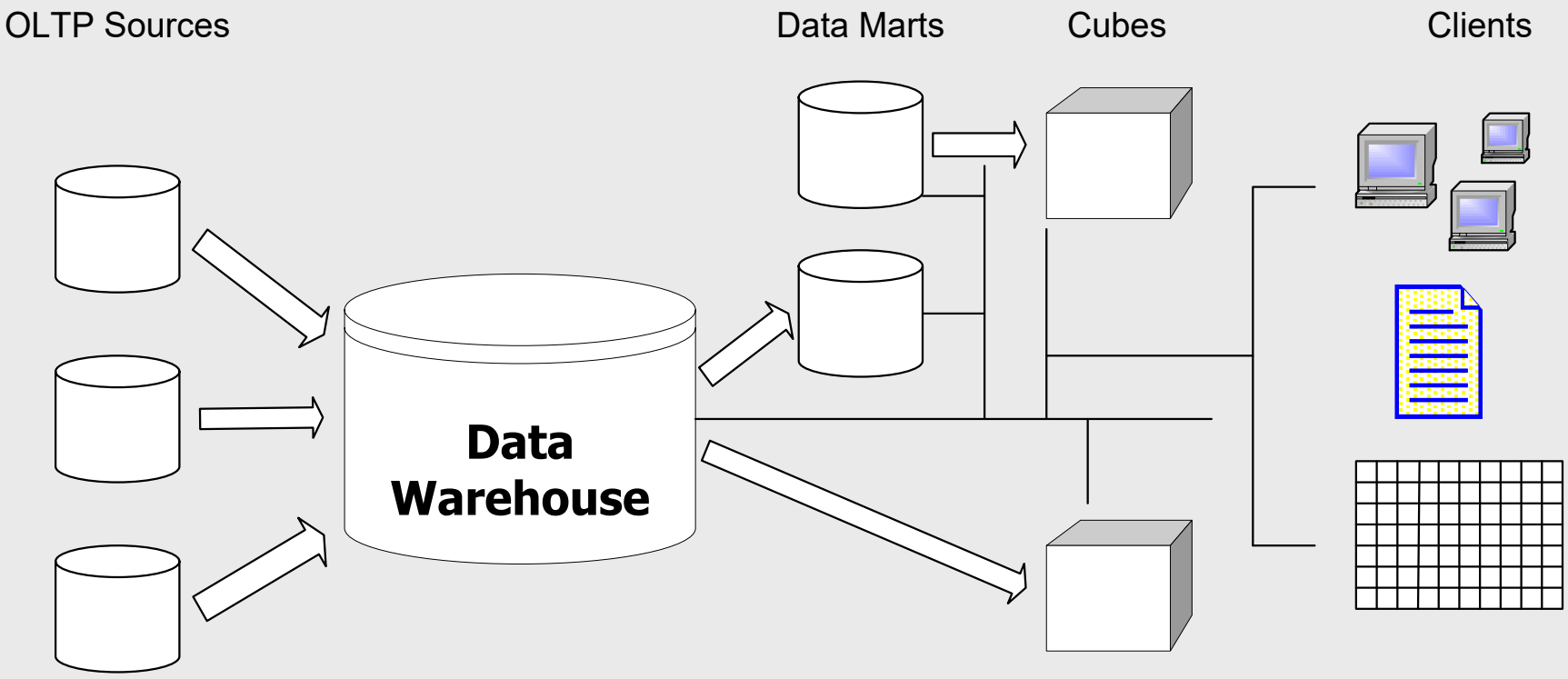
# Goals of Data Warehouse

- **Make an organization's information easily accessible**
- **Present the organization's information consistently**
  - Data must be carefully assembled from a variety of sources around the organization, cleansed quality assured, and released only when it is fit for user consumption
- **Secure and protects information**
- **Serves as a foundation for improved decision making**
  - Have the right data
- **Business users must accept the data warehouse if it is to be useful**

# Advantages of Warehousing Approach

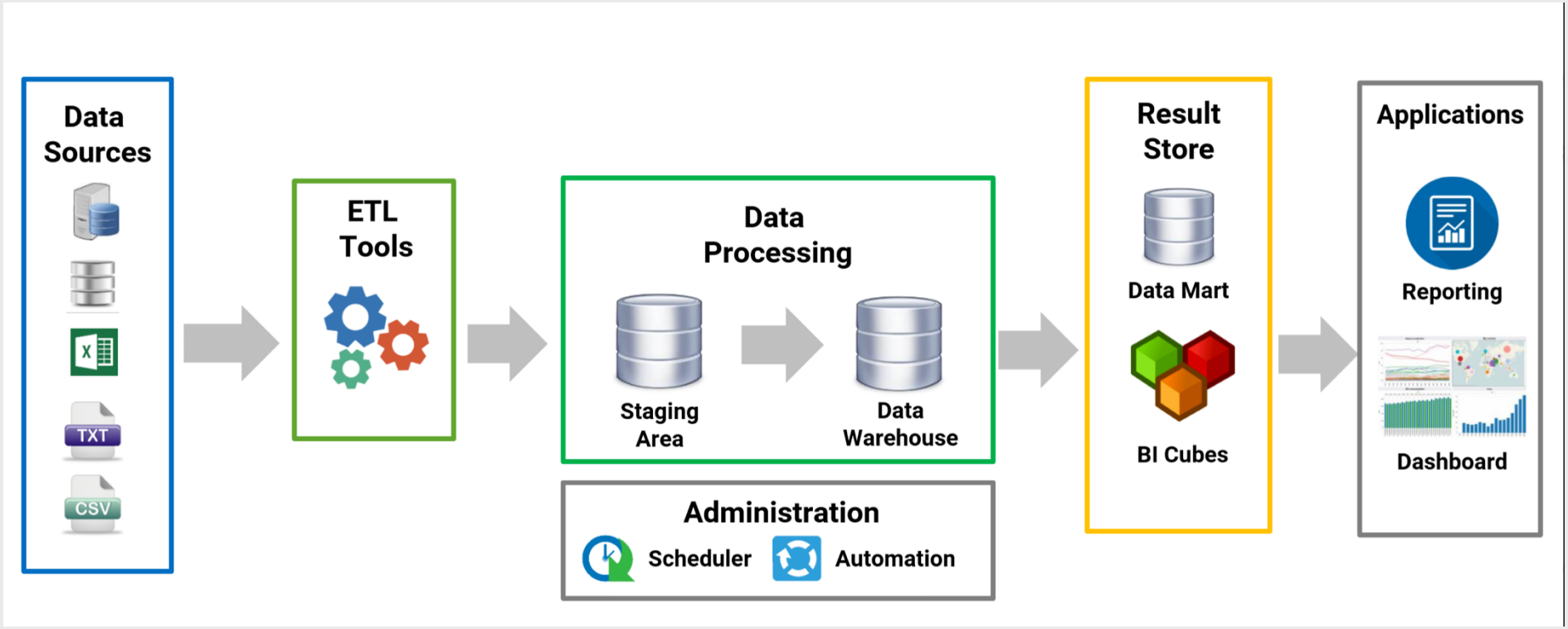
- **High query performance**
- **Doesn't interfere with local processing at sources**
  - **Complex queries** at warehouse
  - **OLTP** (Online Transactional Processing) at operational sources systems
- **At warehouse side**
  - Store historical information
  - Possible to modify, annotate, summarize, restructure, etc.
  - With security concerns

# Basic data warehouse architecture




1. Design the Data Warehouse
2. Populate Data Warehouse
3. Create OLAP Cubes
4. Query Data

# Basic data warehouse architecture



# Users of data warehouse

- **Retail and e-commerce**
  - Analyze and report on sales performance
  - Create machine learning assisted recommendations that drive additional sales
- **Healthcare providers**
  - Can access the most recent insights and use that information to diagnose and treat their patients
- **Transportation providers**
  - Optimize routes, travel times, equipment needs, and staffing requirements
- **Banking and fin-tech**
  - Evaluate risks, detect fraud, and cross-sell services
- **Social media organizations**
  - Need analytic capabilities that can quickly measure ever-changing customer sentiment
- **Governments**
  - Apply business intelligence (BI) to analyze and evaluate citizen-focused programs and assist with policy change decisions

# OLTP(OnLine Transaction Processing) VS OLAP (OnLine Analytical Processing) systems

## OLTP systems

- Mostly updates
- Mainly Pre-defined queries
- Normalized Data Model
- Maintain recent historical data if any
- Must be optimized for writes and small queries.
- Many small transactions
- Based on elementary data

## OLAP systems

- **Mostly Reads**/Read only access
- Mainly **Ad-Hoc queries**
- **De-normalized** Data Model
- **Maintains historical data** for an extended period of time.
- **Must be optimized** for queries involving a large portion of data in the warehouse.
- Queries are **long** and **complex**
- Based on **synthesized data from multiple sources**

# DW for OLAP (On-Line Analytical Processing)

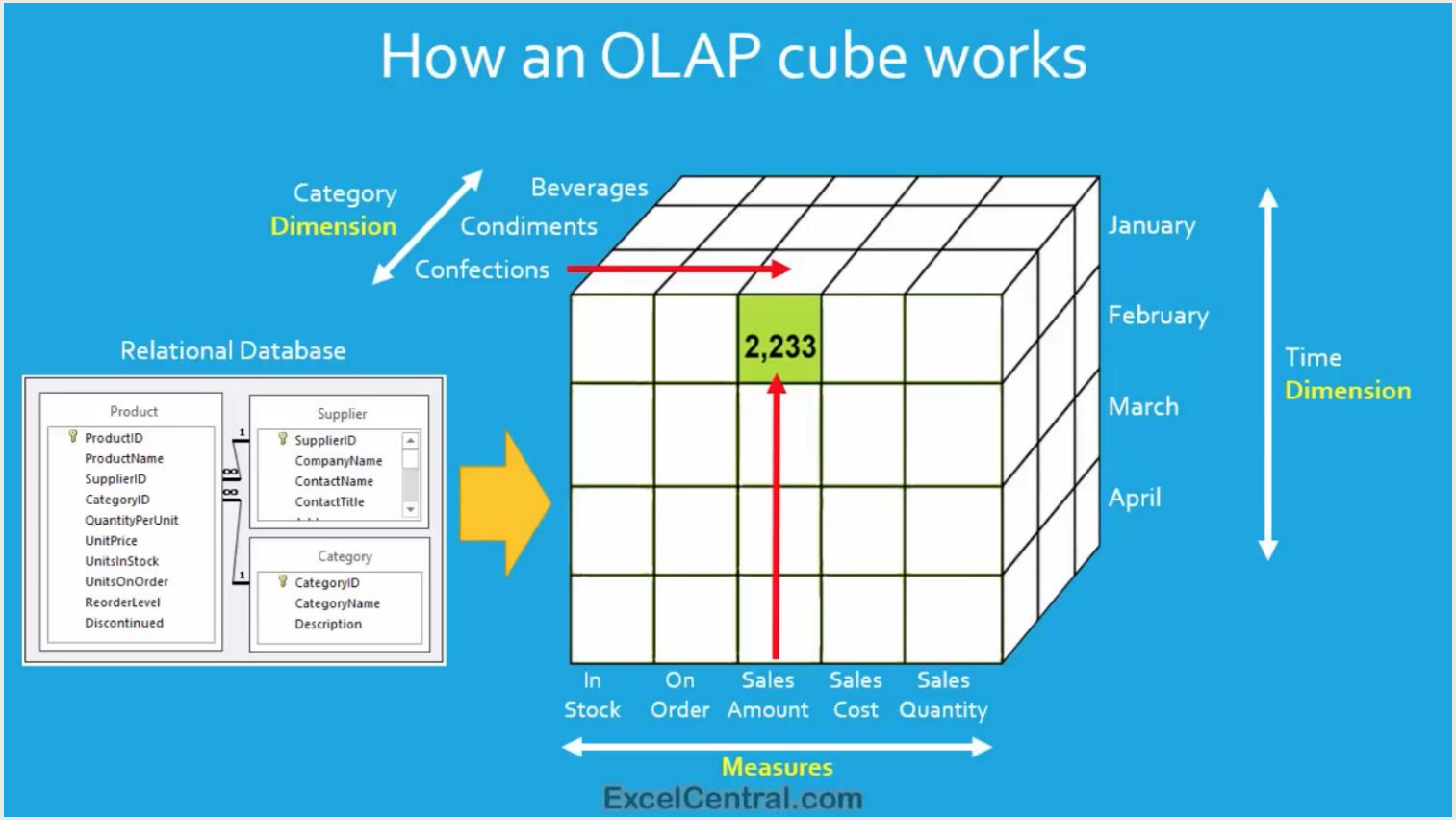
- Difficulty in **analyzing from Large Number of Tables**
  => **Very time-consuming**
- **Instant reporting** using Cube
  => Cube = Set of pre-computed data at specific periods of time

- **Analytics tasks**
  => Aggregation, Prediction, Forecasting, Etc.



# OLAP (On-Line Analytical Processing)

- **Instant reporting using Cube**
  - ⇒ **Cube = Set of pre-computed data at specific periods of time (year, month, week, etc.)**
- **Analytics tasks**
  - ⇒ **Prediction, forecasting, … Etc**



# OLAP (On-Line Analytical Processing)

- **Instant reporting using Cube**
  - ⇒ **Cube = Set of pre-computed data at specific periods of time (year, month, week, etc.)**
- **Analytics tasks**
  - ⇒ **Prediction, forecasting, … Etc**



# DASHBOARDS AND SCORECARDS



# Different Information Worlds

- **Information organization**
  - **Operational systems (Database)**
    - Where the data is put in?
  - **Data warehouse (DW)**
    - Where we got the data out?
- **Data Warehouse users have very different needs than users of operational systems (OLTP).**
- **Users of operational system**
  - Turn the wheels of an organization
  - Almost always deal with one record at a time
  - Repeatedly perform the same operational tasks over and over
- **Users of data warehouse**
  - Watch the wheels of the organization turn
  - Questions often require that hundreds or thousands of rows
  - Continuously change the kinds of questions they ask

  => **Ad-hoc queries**

# A Data Warehouse is...

- A **subject-oriented**, **integrated**, **time-variant** and **non-volatile collection of data** in support of management's decision-making process.
- Stored collection of diverse and integrated data
- **Subject-oriented**
  - Organized by subject, not by application
  - Used for decision-support, analysis, etc.
- Optimized differently from transaction-oriented or operational databases.
- User interface aimed at target-users.
- **Time-variant**
  - Historical
  - Time attributes are important
- **Non-volatile**
  - Updates infrequent
  - May be append-only

# Four Key Characteristics

- **Subject-Oriented:**
  - Organized by business subjects (sales, customers, products).
- **Integrated:**
  - Data from multiple sources combined consistently
- **Time-Variant:**
  - Historical data preserved with time stamps.
- **Non-Volatile:**
  - Data is stable
  - Updates are infrequent

# Data warehouse design & implementation steps

1. **Requirement Analysis**
   - Identify business factors !
   - Gather & analyze data sources
   - Select the business process to
2. **Multi-Dimensional modeling**
3. **Declare the grain of the business process**
4. **Design the dimensions**
5. **Design the facts**
6. **Data warehouse implementation**

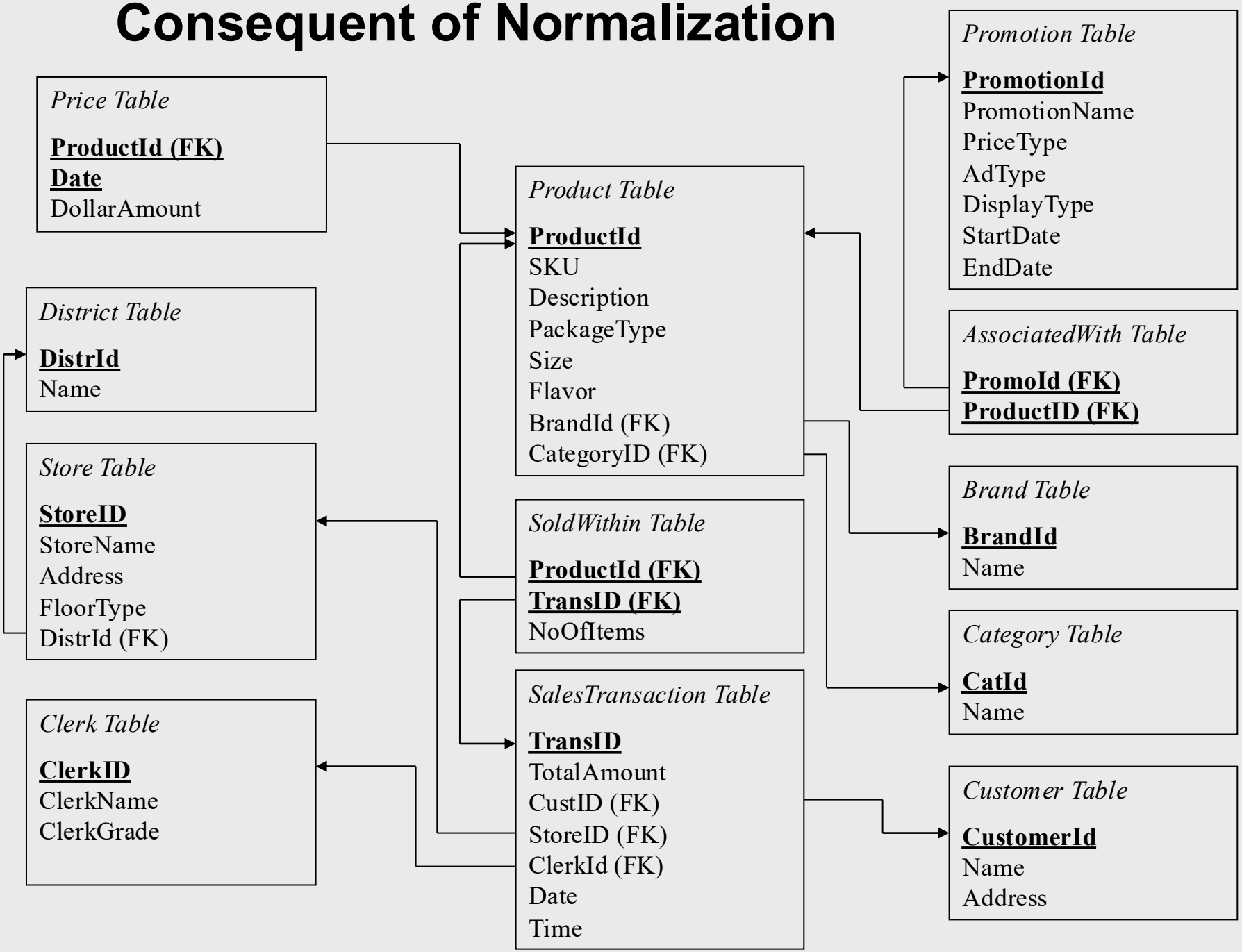
# Data Warehouse Design

- **OLTP Database Design**
  - UML Modeling + Normalization
  - Design  technique that seeks to eliminate data redundancy to quick insert, delete and update operations
  - Beneficial to transaction processing in relational databases

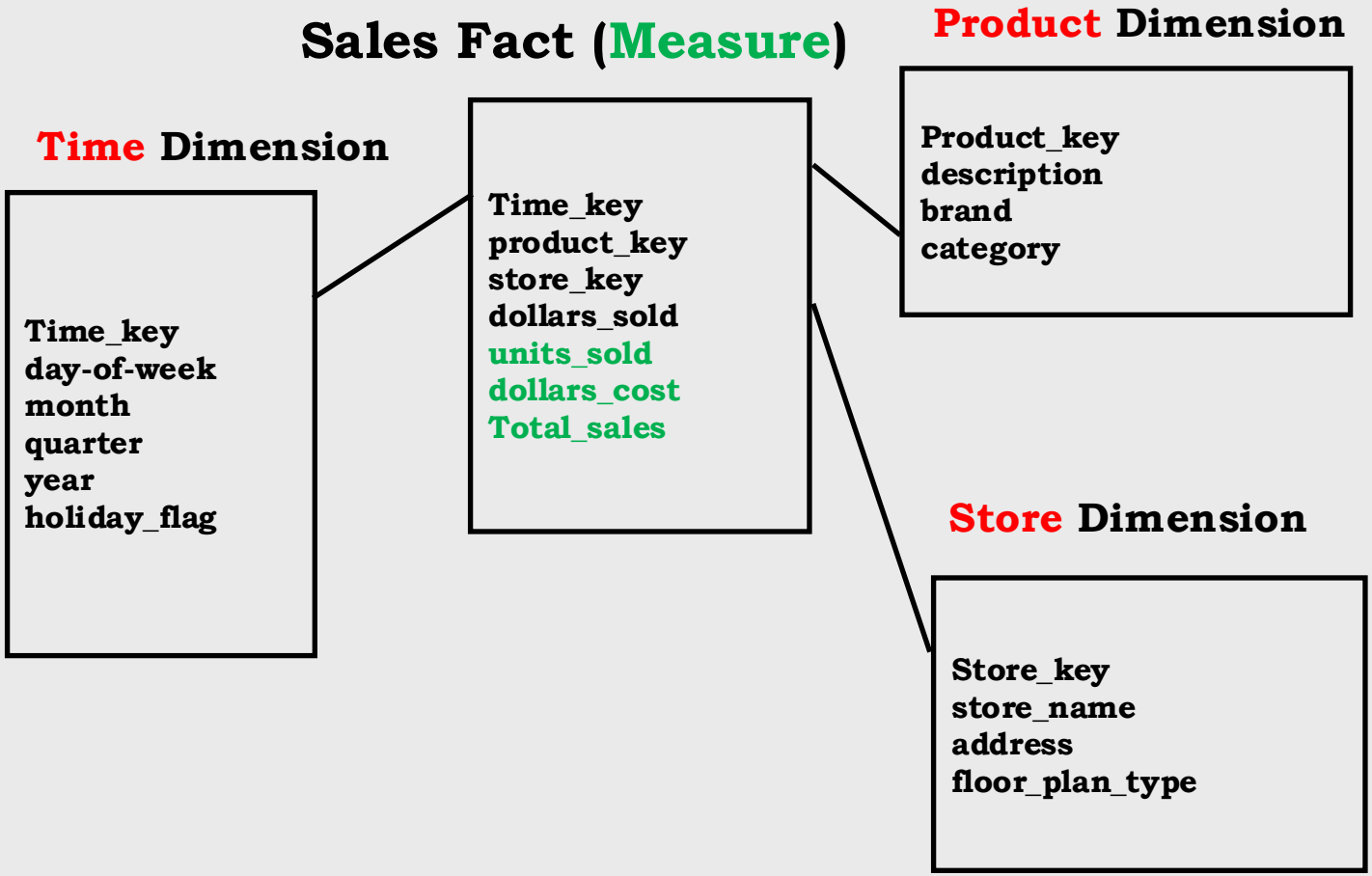
- **Data Warehouse Design**
  - Design  technique used for designing data warehouses

  **=> Multi-Dimensional Modeling** (De-Normalization)

# Consequent of Normalization



# Dimensional Modelling Example: Star-Schema



# UML-Modeling vs. Dimensional Modeling

- **Dimensional Model**
  - represent subject areas within the enterprise and serves as a blueprint for **OLAP analytical systems**

- **UML-Model**
  - represents business processes within the enterprise and serves as a blueprint for **OLTP operational database systems**

# Dimensional Modeling

- **Dimensional Modeling gives us a way to visualize and analyze data.**
- **The CEO's perspective:**
  - **"We sell products in various stores, and we measure our sale figure over time."**
- **From the data warehouse designer's perspective, we hear three dimensions:**
  - **We sell *Products***
  - **in various *Stores***
  - **and measure performance over *time*.**

# Dimensional Modeling - contd.

- Management may be interested in examining sales figures in a certain city by product, by time period, by salesperson, and by store.
- The more dimensions involved, the more difficult it is to represent in a single table.

# Dimensional Modeling

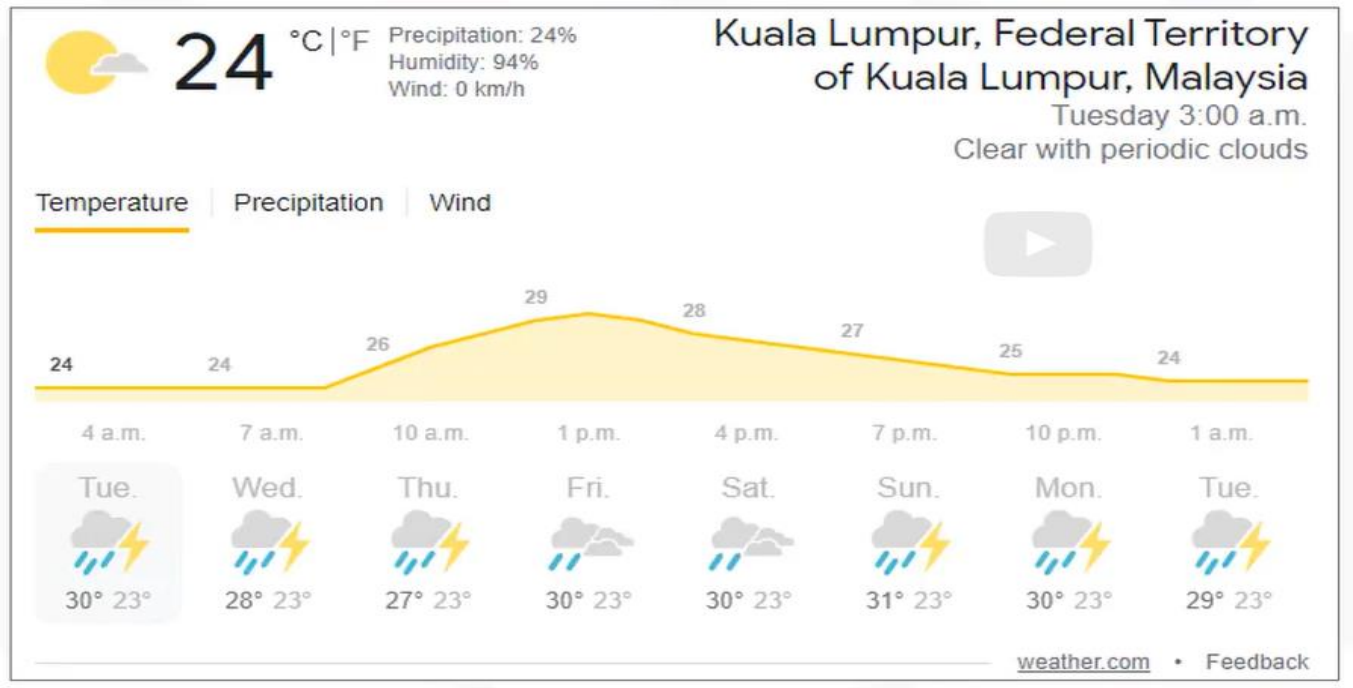
- **Fact Tables**
- **Dimension Tables**

# Dimensional Model

- **FACTS (Measures)**
  - **Fact are (usually numerical) measures of business**
  - **Values that can be measured and aggregated (mostly)**
    **Examples**: Sales amount, quantity sold, profit
  - **Stored in the fact table (center of star schema)**
  - **Fact table is the largest table in the star schema and is composed of large volumes of data**
- **DIMENSIONS**
  - **Textual descriptions of the business**
  - **Descriptive attributes that provide context to facts**
    **Examples**: Date, Product, Customer, Store, Employee
  - **Stored in dimension tables**
  - **Dimension tables contain relatively small amounts of relatively static data**
  - **Dimension table: usually not-normalized**
  - **Contain large number of attributes for analysis ability**

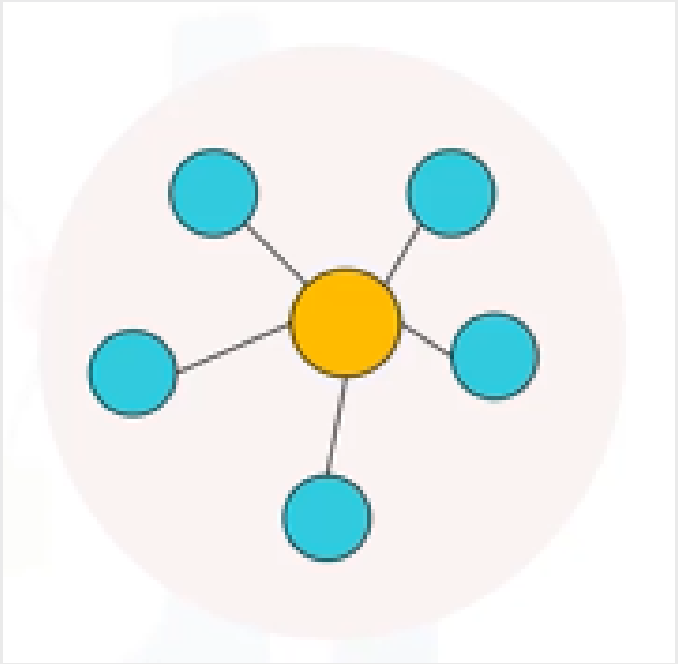
# Facts and Dimensions

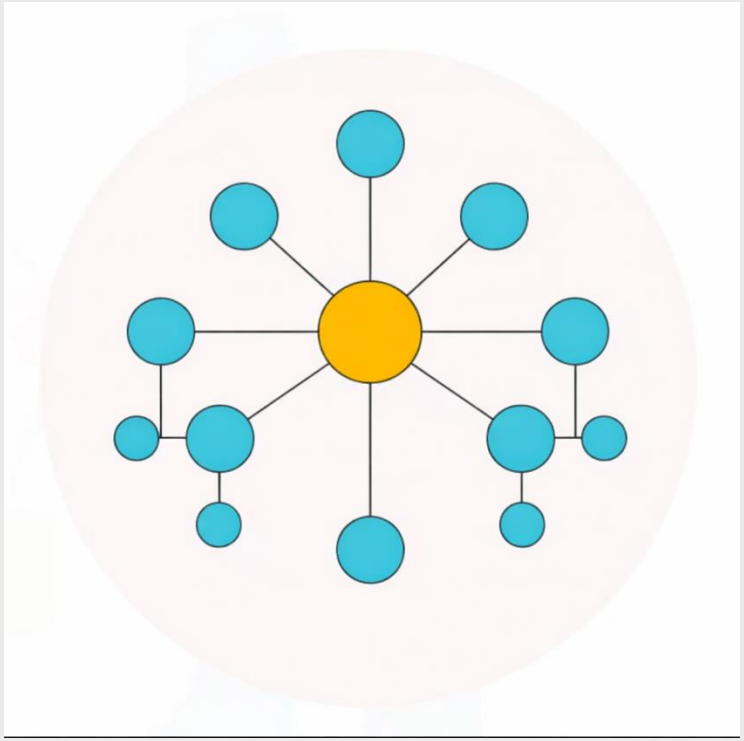
- Data can be organized as: Facts (Measures) and dimensions.
- Facts are usually quantities (or qualitative) which can be measured, such as
  - Sale figures,
  - Temperature
- Dimensions are attributes which are related to facts.
- Dimensions provide context to facts



# Star-schema: Facts and Dimensions

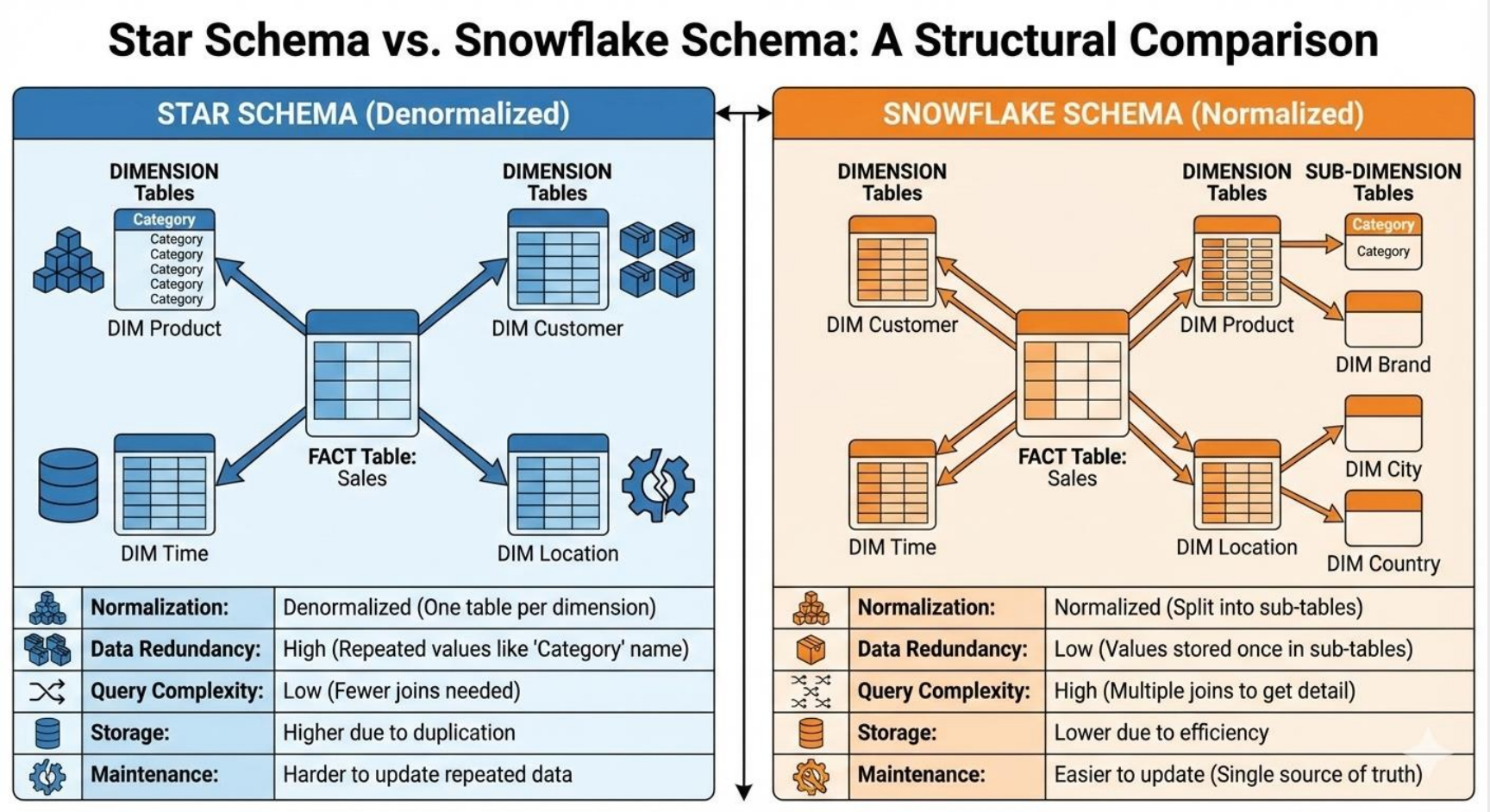
- **Star-schema** is a graph whose edges are relations between facts and dimensions (mostly un-normalized)
- **Keys** are used to connect fact(s) with dimension(s)
- **Snow-flake schemas** are **normalized star-schemas**
  - Dimensions tables split into child tables
  - Don't need to be fully normalized





# Star-schema VS Snowflake-schema

- **Snow-flake schemas** are **normalized star-schemas**
  - Dimensions tables split into child tables
  - Don’t need to be fully normalized



# Dimensional Modeling Vocabulary

- **Fact Table**
  - **Is the primary table in a dimensional model**
  - **Facts are numeric measurements (values) that represent a specific business aspect or activity**

| Daily Sales Fact Table |
| --- |
| **Date Key(FK)** |
| **Product Key(FK)** |
| **Store Key(FK)** |
| **Quantity Sold** |
| **Dollor Sales** |
| **Amount** |

# Fact Table

- **Fact Table is where numerical measurements of the business are stored.**
- **Each measurement is taken at the intersection of all the dimensions.**
- **For every query made against the fact table may use hundreds of thousands of individual records to construct an answer set.**

# Dimensional Modeling Vocabulary

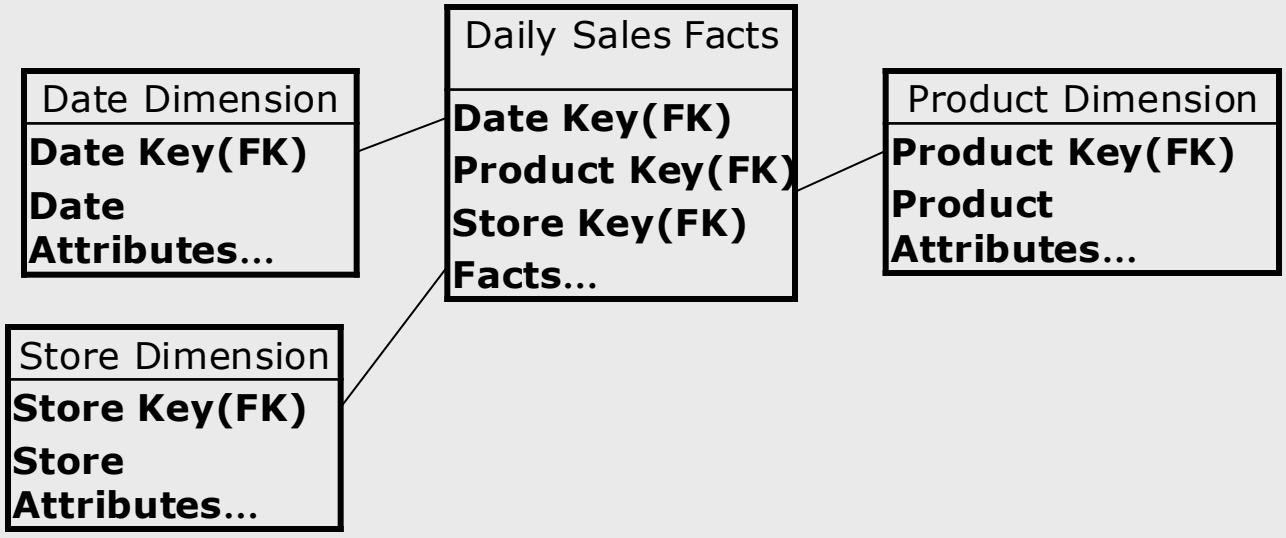
- **Dimension Tables**
  - **Are integral companions to a fact table**
  - **Contain the textual descriptors of the business**
  - **Have many columns or attributes**

| Product Dimension Table |
| --- |
| **Product Key(PK)** |
| **Product Description** |
| **SKU Number(Natural key)** |
| **Brand Description** |
| **Category Description** |
| **Department Description** |
| **Package Type Description** |
| **Package Size** |
| **Fat Content Description** |
| **Diet Type Description** |
| **Weight** |
| **Weight Units of Measure** |
| **Storage Type** |
| **Shelf Life Type** |
| **Shelf Width** |
| **Shelf Height** |
| **Shelf Depth** |
| **... and many more** |

Sample dimension Table

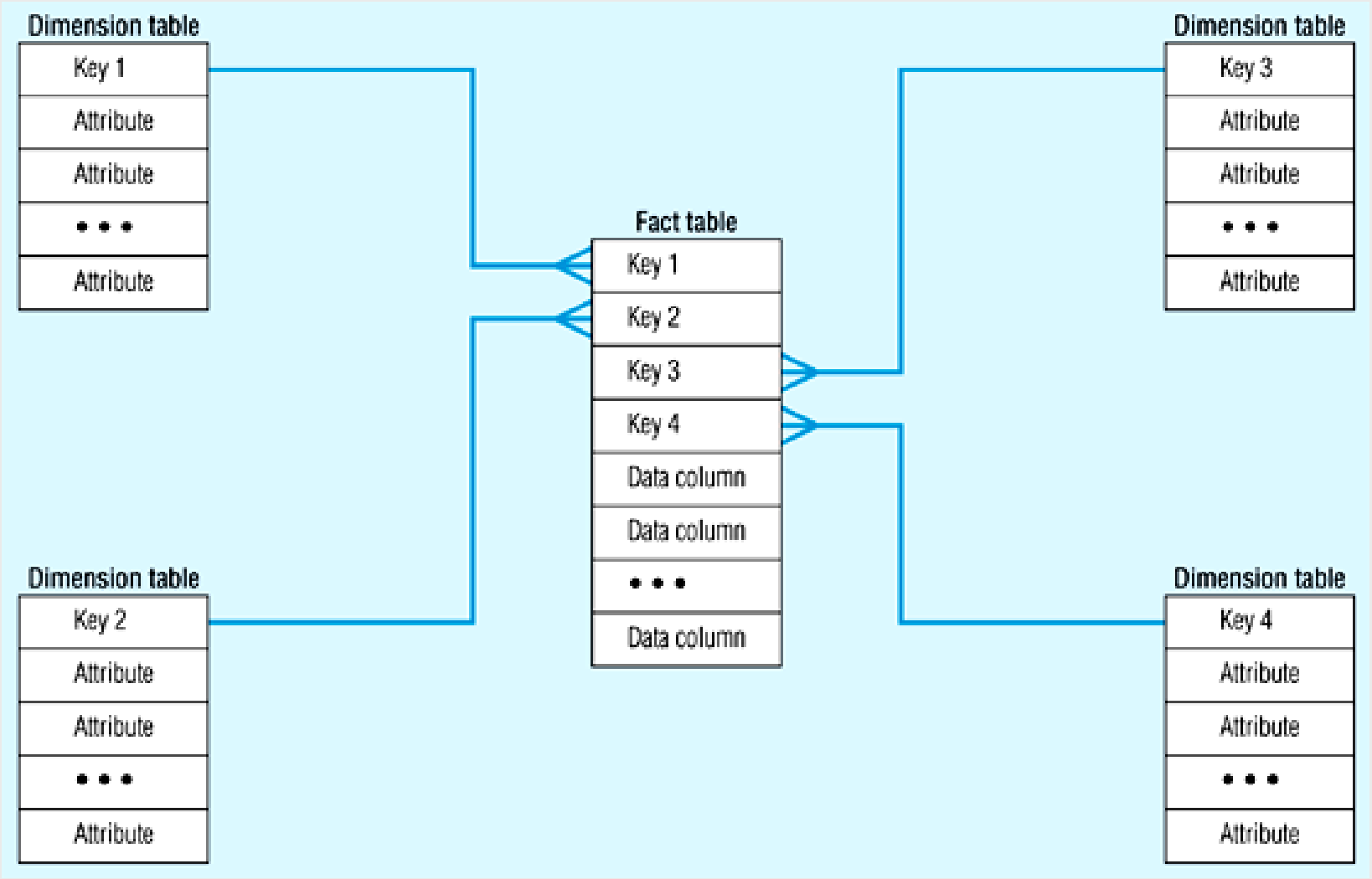
# Dimensional Modeling Vocabulary

- **Bringing Together Facts and Dimensions**
  - The fact table consisting of numeric measurements is joined to a set of dimension tables filled with descriptive attributes
  - This characteristic star-like structure is often called a **star-schema**
  - All dimension are symmetrically equal entry points into the fact table
    - No preferences for any query

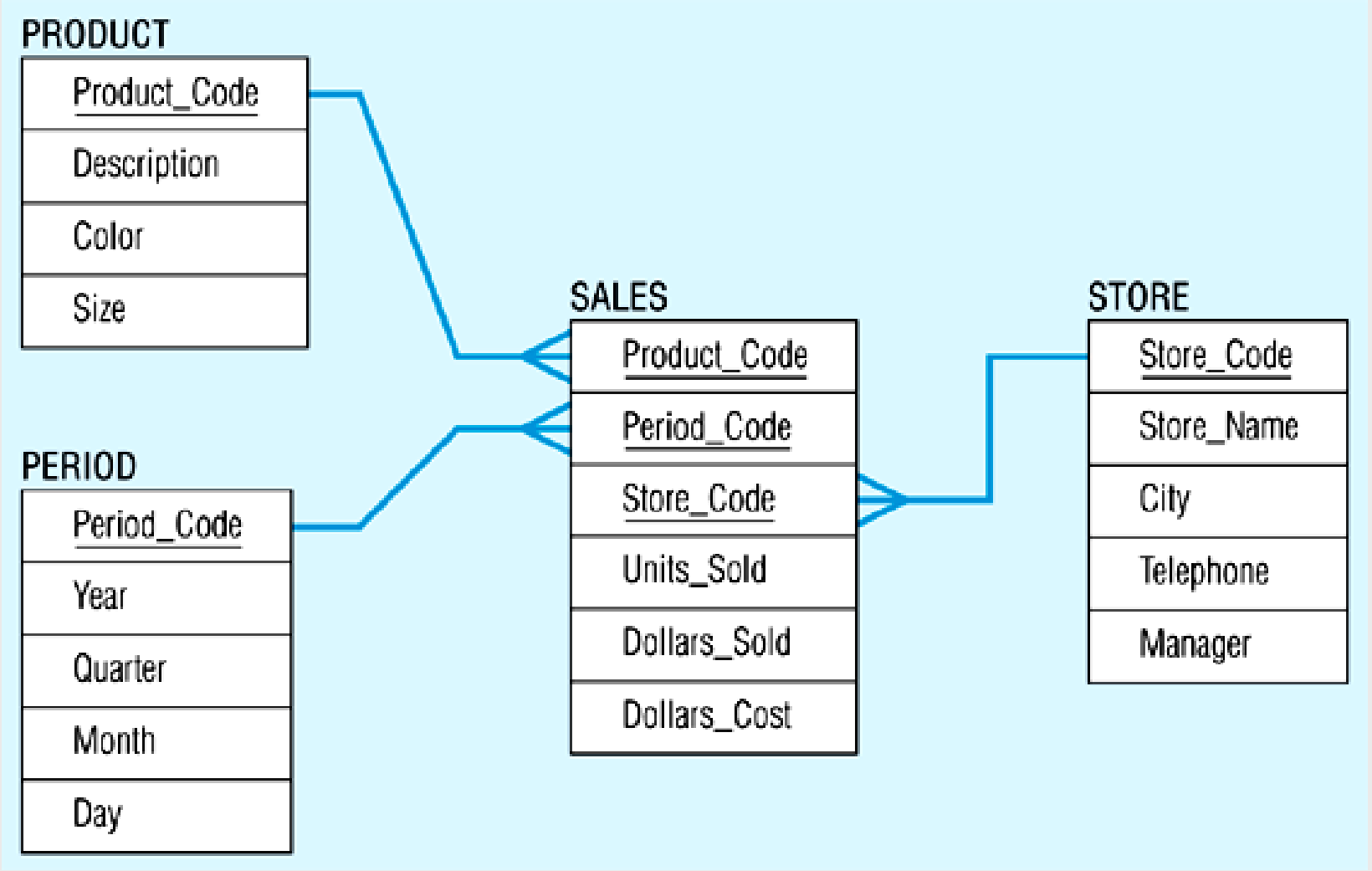



Fact and dimension Tables in a dimensional model

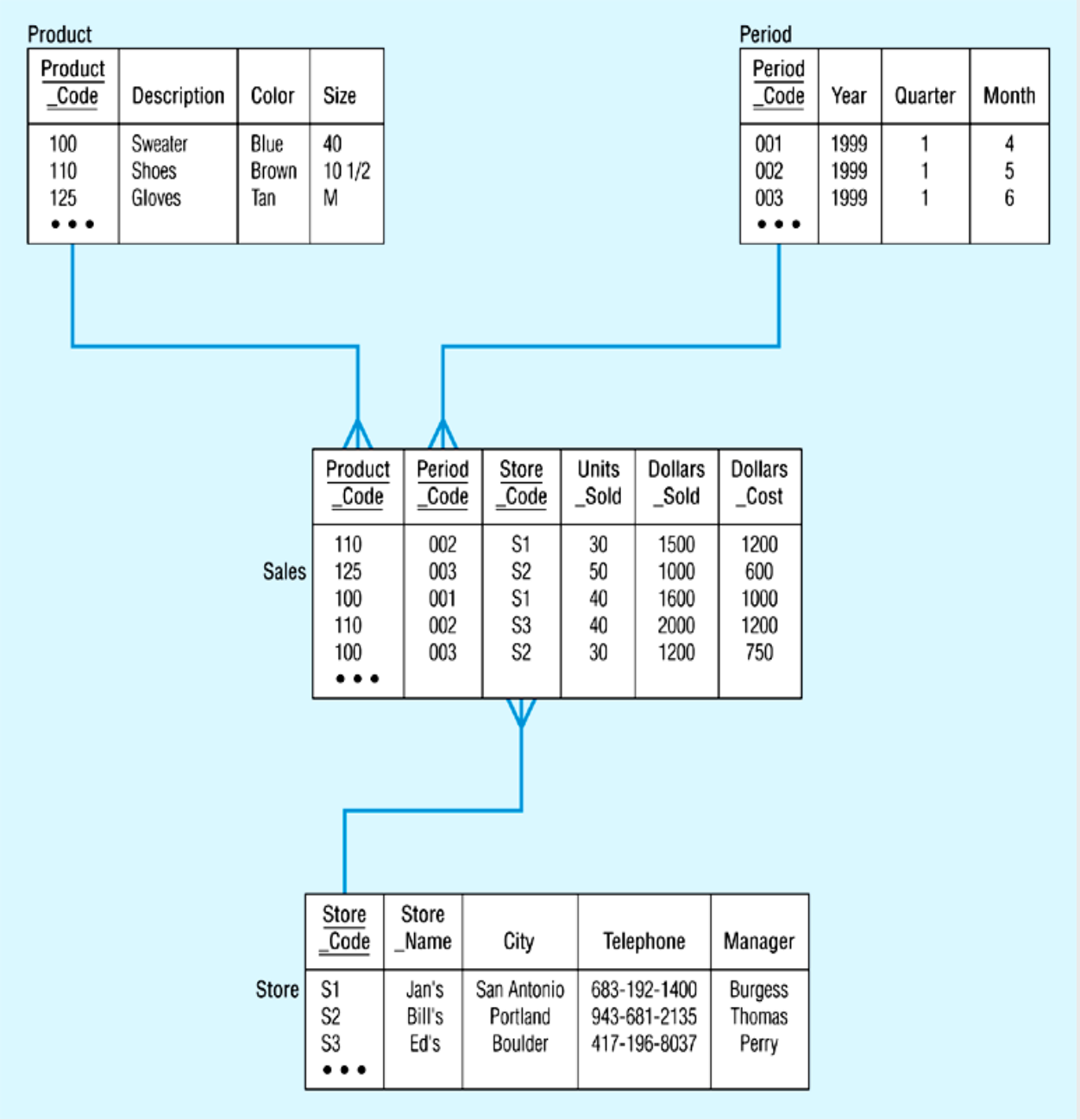
# Star-Schema


Dimension table
Key 1
Attribute
Attribute
• • •
Attribute
Dimension table
Key 3
Attribute
Attribute
• • •
Attribute
Fact table
Key 1
Key 2
Key 3
Key 4
Data column
Data column
• • •
Data column
Dimension table
Key 2
Attribute
Attribute
• • •
Attribute
Dimension table
Key 4
Attribute
Attribute
• • •
Attribute

# Star-Schema Example



# Star-Schema with Sample Data



# Fact Table Components

## Dimension Tables

*customer_dim*

| | | |
|---|---|---|
| 201 | ALFI | Alfreds |
| | | |

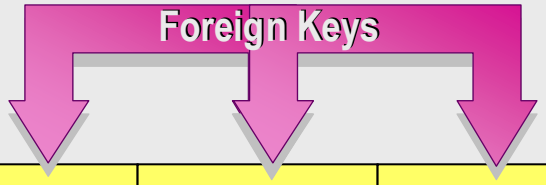
*product_dim*

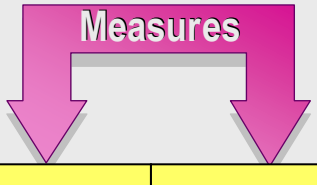
| | | |
|---|---|---|
| 25 | 123 | Chai |
| | | |

*time_dim*

| | |
|---|---|
| 134 | 1/1/2000 |
| | |

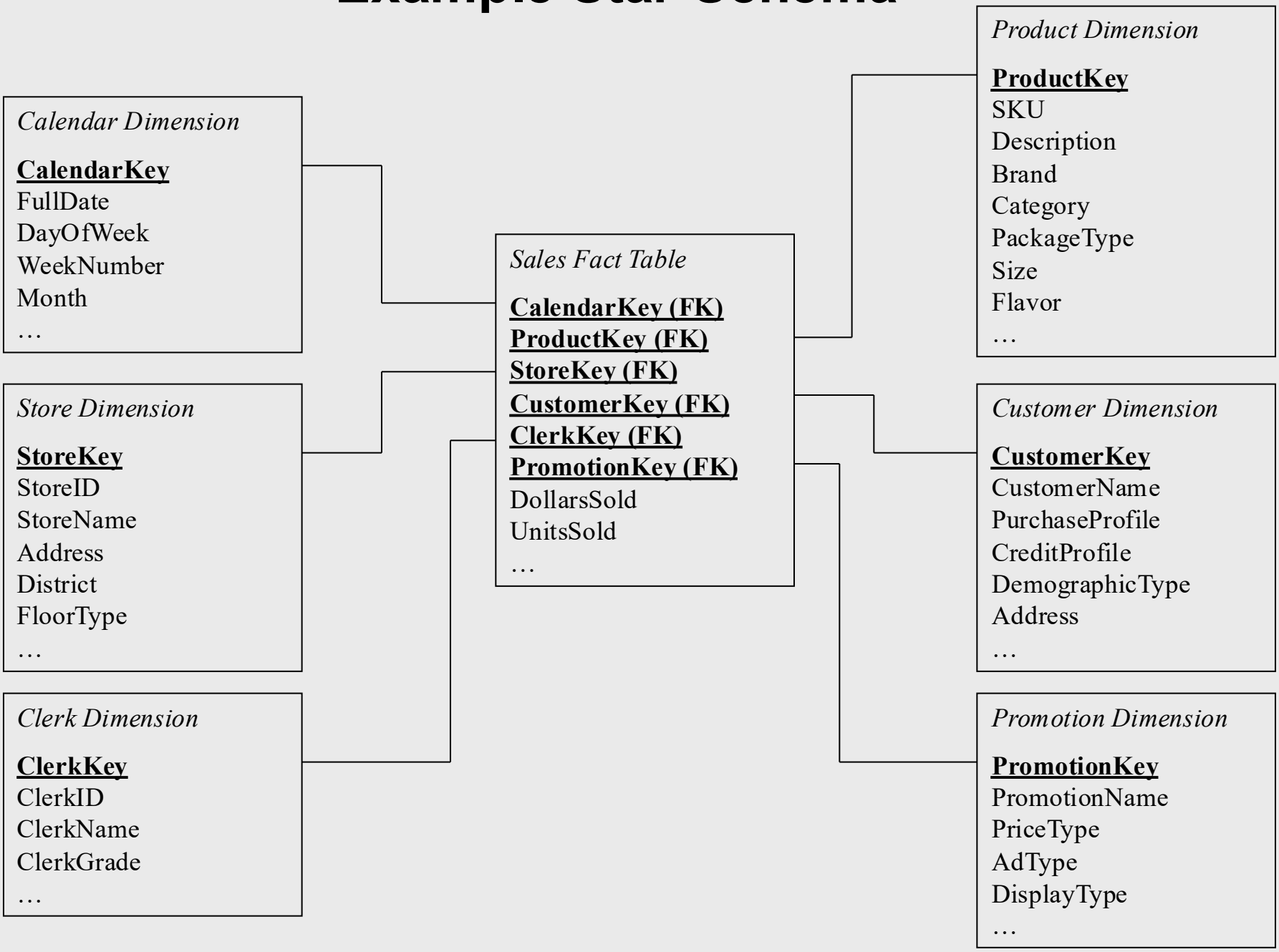
## sales_fact Table



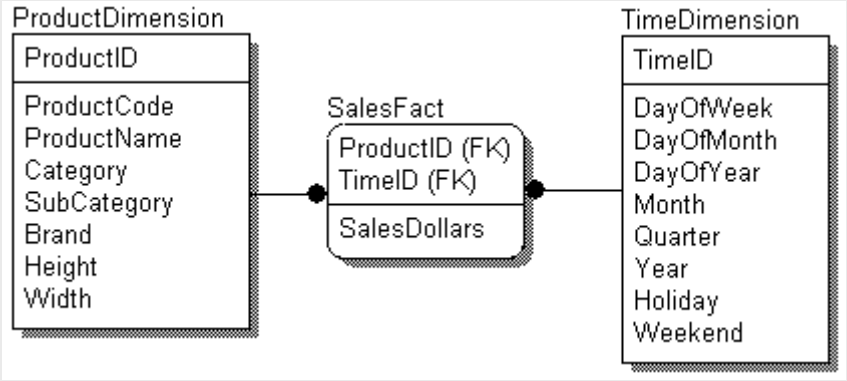

| *customer_key* | *product_key* | *time_key* | *quantity_sales* | *amount_sales* |
|---|---|---|---|---|
| 201 | 25 | 134 | 400 | 10,789 |
| | | | | |

# Example Star-Schema



# Example of Using Star-schema

- A simplified example of
- 2 dimension tables joined to the fact table.
- The fact table has only one measure: SalesDollars.




**SQL: Total sales** for a **particular month** for a **particular category**

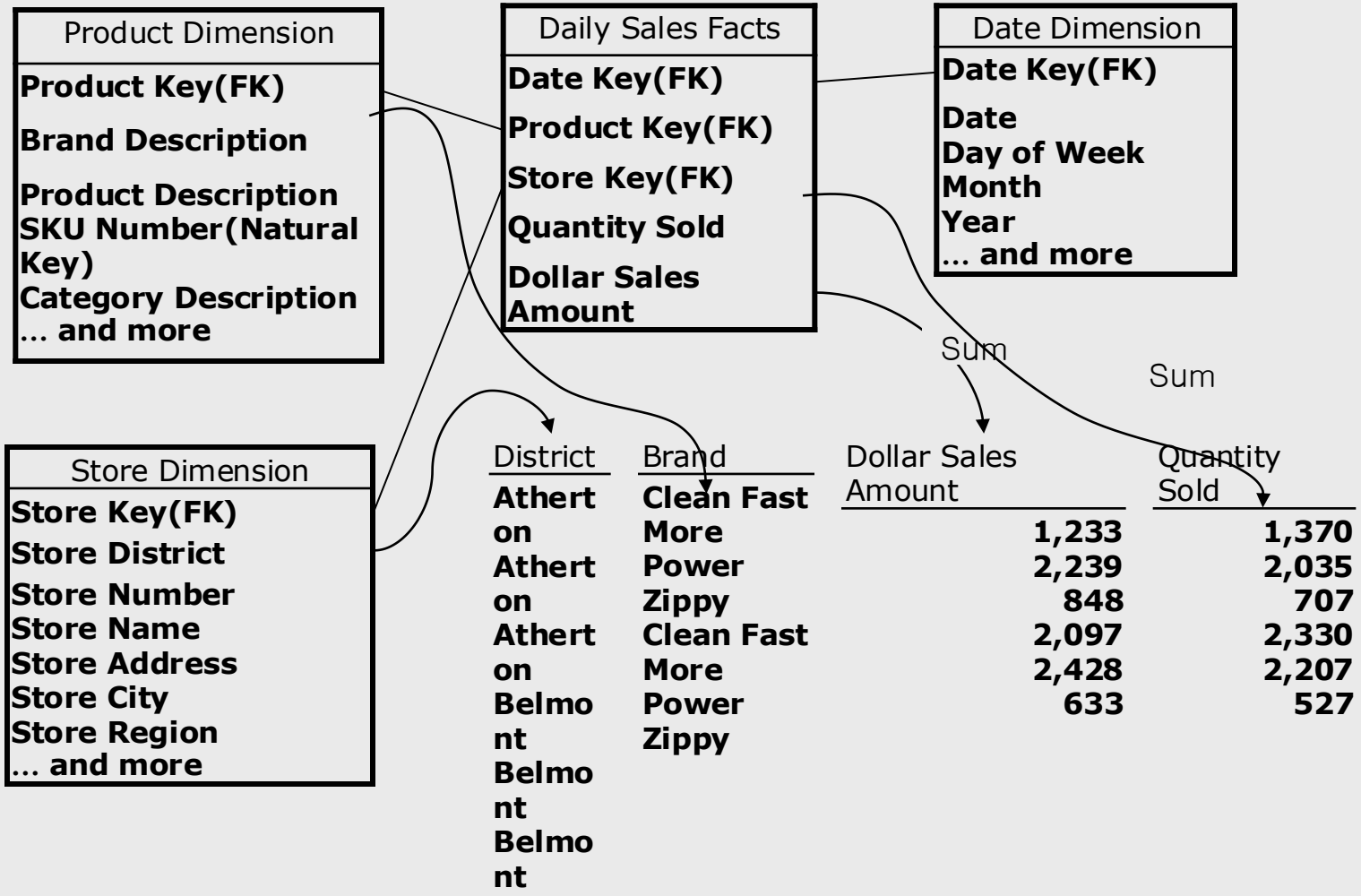
```
SELECT    Sum ( SalesFact.SalesDollars) AS SumOfSalesDollars
FROM      TimeDimension JOIN (ProductDimension JOIN
          SalesFact ON Product Dimension.ProductID = SalesFact.ProductID)
          ON TimeDimension.TimeID = SalesFact.TimeID
WHERE     ProductDimension.Category='Brass Goods' AND TimeDimension.Month=3
          AND TimeDimension.Year=1999
```

**Drill-down** to a **sub-category**:

```
SELECT    Sum ( SalesFact.SalesDollars) AS SumOfSalesDollars
FROM      TimeDimension JOIN (ProductDimension JOIN
          SalesFact ON Product Dimension.ProductID = SalesFact.ProductID)
          ON TimeDimension.TimeID = SalesFact.TimeID
WHERE     ProductDimension.Subcategory='Widgets' AND TimeDimension.Month=3
          AND TimeDimension.Year=1999
```

# Dimensional Modeling Vocabulary

- Bringing Together Facts and Dimensions
- **Dragging and dropping dimensional attributes and facts into simple report**



# Data warehouse design 4-steps

1. **Identify Business Process via Requirement Analysis**
   - Identify business factors !
   - Gather & analyze data sources
   - Select the **business process** to
2. **Declare the granularity (grain) of the business process**
3. **Determine the dimensions**
4. **Determine the facts**

# The 4-Step Dimensional Design Process

- **Step 1: Select the Business Process**
  - What business activity are we analyzing?
    - Sales transactions analysis,
    - Order processing,
    - Inventory management,
    - Customer service calls analysis,
    - Website clicks
- **Step 2: Declare the Grain**
  - What does each row in the fact table represent?
    - Individual line item on a sales receipt,
    - Daily sales summary by product and store,
    - Monthly customer account balance,
    - Individual web page click

# The 4-Step Dimensional Design Process

- **Step 3: Choose the Dimensions**
  - How do we want to analyze the facts?
  - What are the different ways we want to analyze the facts?
  - **Ask yourself:**
    - How do business users want to filter the data?
    - What attributes do they use to group and sort?
    - What provides context to the measurements?
  - **Very Common dimensions:**
    - Date/Time: When did it happen?
    - Product: What was involved?
    - Customer: Who was involved?
    - Location: Where did it happen?
    - Employee: Who handled it?

# The 4-Step Dimensional Design Process

- **Step 4: Identify the Facts**
  - What are we measuring?
  - **Facts should be:**
    - Numeric (usually)
    - Additive (can be summed across dimensions)
    - Meaningful for business analysis
  - **Types of facts:**
    - **Additive**: Can be summed across all dimensions (sales amount, quantity)
    - **Semi-additive**: Can be summed across some dimensions (ex. account balance)
    - **Non-additive**: Cannot be summed (ratios, percentages)

# Four-Step Dimensional Design Process

- **Initial definitions**
  1. **Select the business process to model**

     **Example of business process**

     **:raw materials purchasing, orders, shipments, invoicing, inventory**
  2. **Declare the grain of the business process**

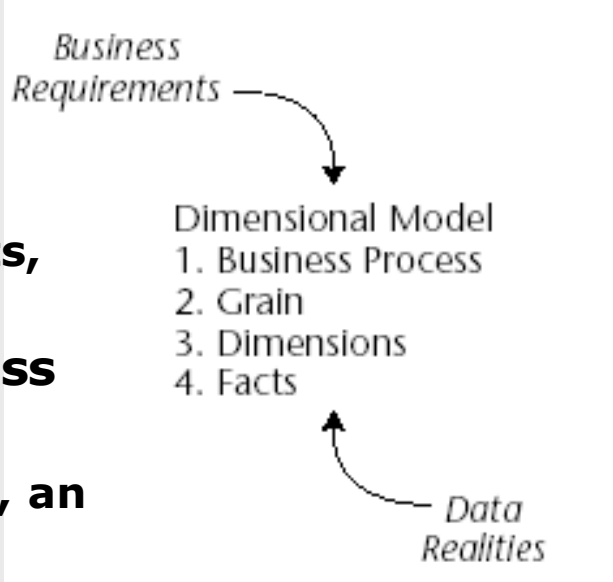
     **Example of grain declarations**

     **:an line item on a bill received from a doctor, an individual boarding pass to get on a flight**
  3. **Choose the dimensions that apply to each fact table row**

     **Example of dimensions**

     **:date, product, customer, transaction type, status**
  4. **Identify the numeric facts that will populate each fact table row**

     **"What are we measuring?" the answer is used to determine the facts**

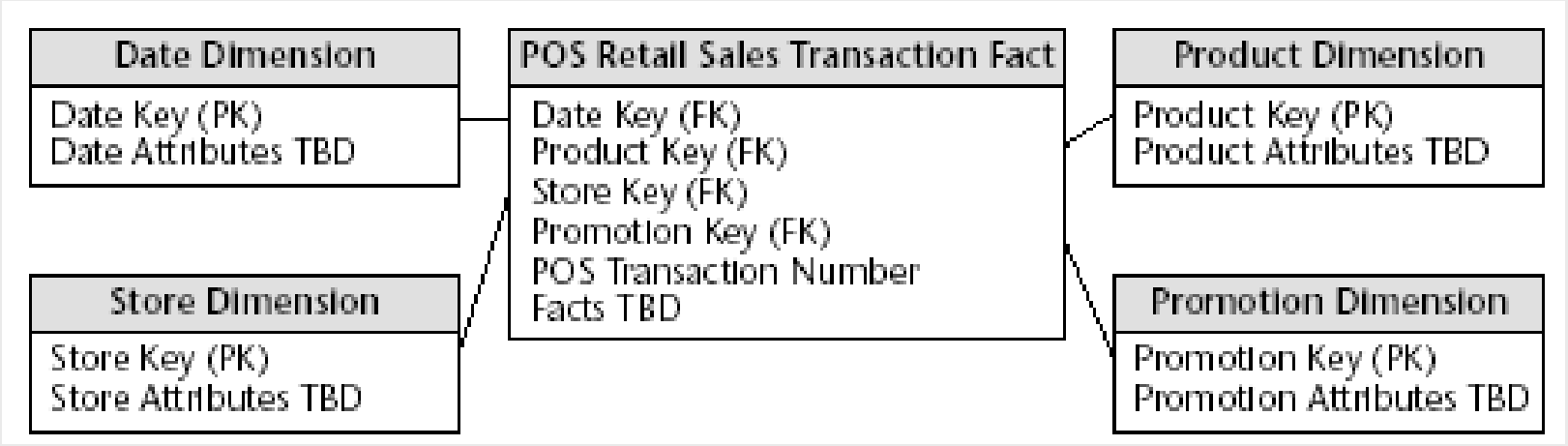


# Retail Case Study

- **Dimensional Design Process**
  - **Step1. Select the business process**
    - Sale Analysis
      - "POS retail sales" business process to analyze
      - what products are selling in which stores
      - on what days
      - under what promotional conditions
      - How much sale amount pert transaction
  - **Step2. Declare the grain**
    - Example, the most granular data is an **individual line item on a POS transaction**
  - **Step3. Choose the dimensions**
    - Once the grain the fact table has been chosen,
      - The **date**, **product**, and **store** primary dimensions fall out immediately
    - It is possible to add more dimensions to the basic grain of the fact table
      - We can ask whether other dimensions can be attributed to the data, such as the **promotion** under which the product is sold

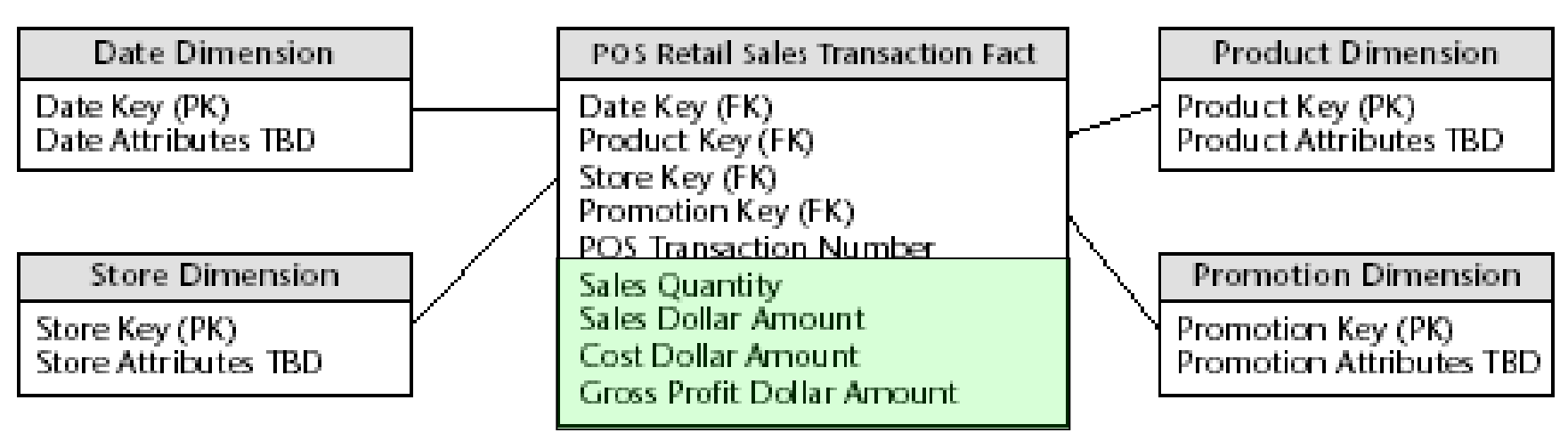
# Retail Case Study

- **Dimensional Design Process**
  - **Step3. Choose the dimensions**
    - Fig 2.2 Preliminary retail sales schema



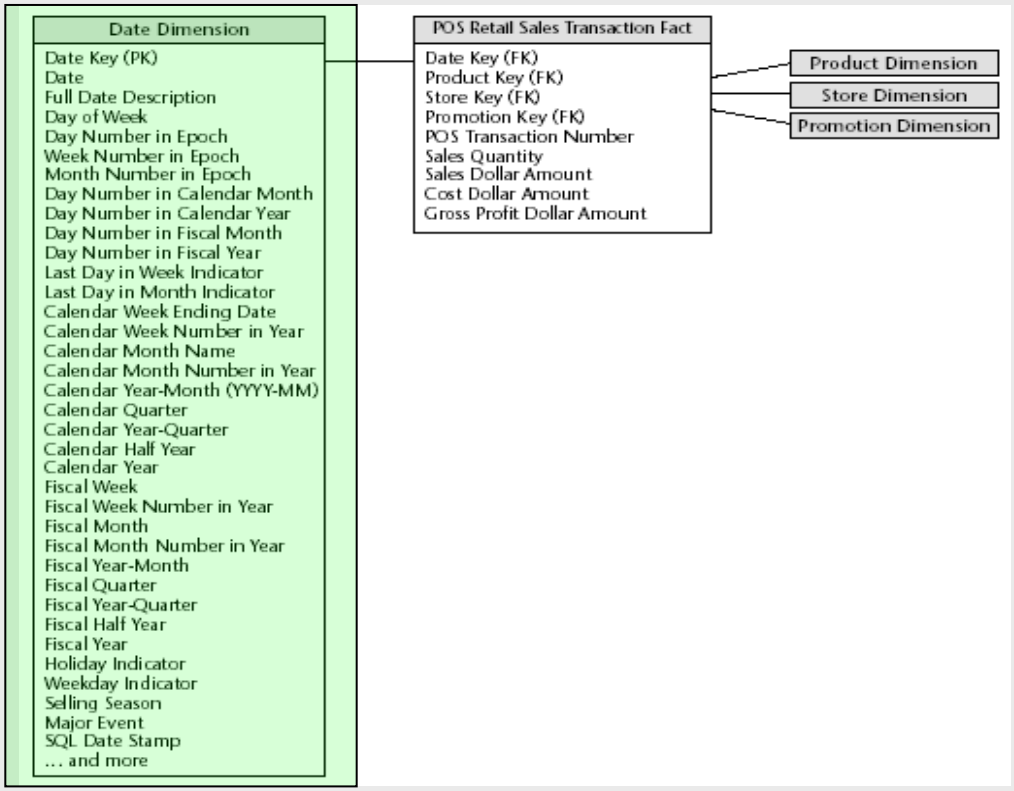
# Retail Case Study

- **Dimensional Design Process**
  - **Step4. Identify the facts (measures)**



# Dimension Table Attributes

- Focus on filling the dimension tables with **robust attributes**
- **Date** Dimension
  - is the one dimension nearly **guaranteed to be in every data warehouse**
    - because data mostly time series
  - Date dimension in the retail sales schema



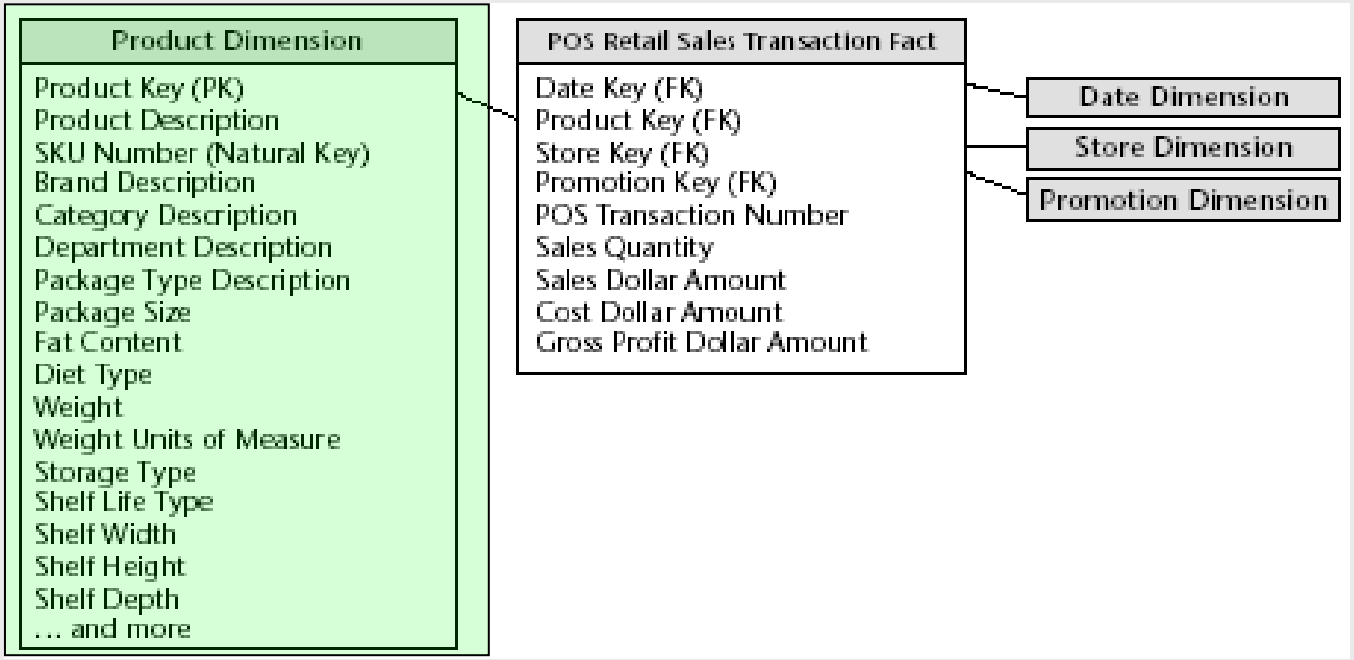
# Dimension Table Attributes

- **Date Dimension**
  - **Example content of a Date dimension table**

| Date Key | Date | Full Date Description | Day of Week | Calendar Month | Calendar Year | Fiscal Year-Month | Holiday Indicator | Weekday Indicator |
|---|---|---|---|---|---|---|---|---|
| 1 | 01/01/2002 | January 1, 2002 | Tuesday | January | 2002 | F2002-01 | Holiday | Weekday |
| 2 | 01/02/2002 | January 2, 2002 | Wednesday | January | 2002 | F2002-01 | Non-Holiday | Weekday |
| 3 | 01/03/2002 | January 3, 2002 | Thursday | January | 2002 | F2002-01 | Non-Holiday | Weekday |
| 4 | 01/04/2002 | January 4, 2002 | Friday | January | 2002 | F2002-01 | Non-Holiday | Weekday |
| 5 | 01/05/2002 | January 5, 2002 | Saturday | January | 2002 | F2002-01 | Non-Holiday | Weekend |
| 6 | 01/06/2002 | January 6, 2002 | Sunday | January | 2002 | F2002-01 | Non-Holiday | Weekend |
| 7 | 01/07/2002 | January 7, 2002 | Monday | January | 2002 | F2002-01 | Non-Holiday | Weekday |
| 8 | 01/08/2002 | January 8, 2002 | Tuesday | January | 2002 | F2002-01 | Non-Holiday | Weekday |

# Dimension Table Attributes

- **Product Dimension**
  - **Describes every stock keeping unit (SKU) in the grocery store**
  - **Product dimension in the retail sales schema**



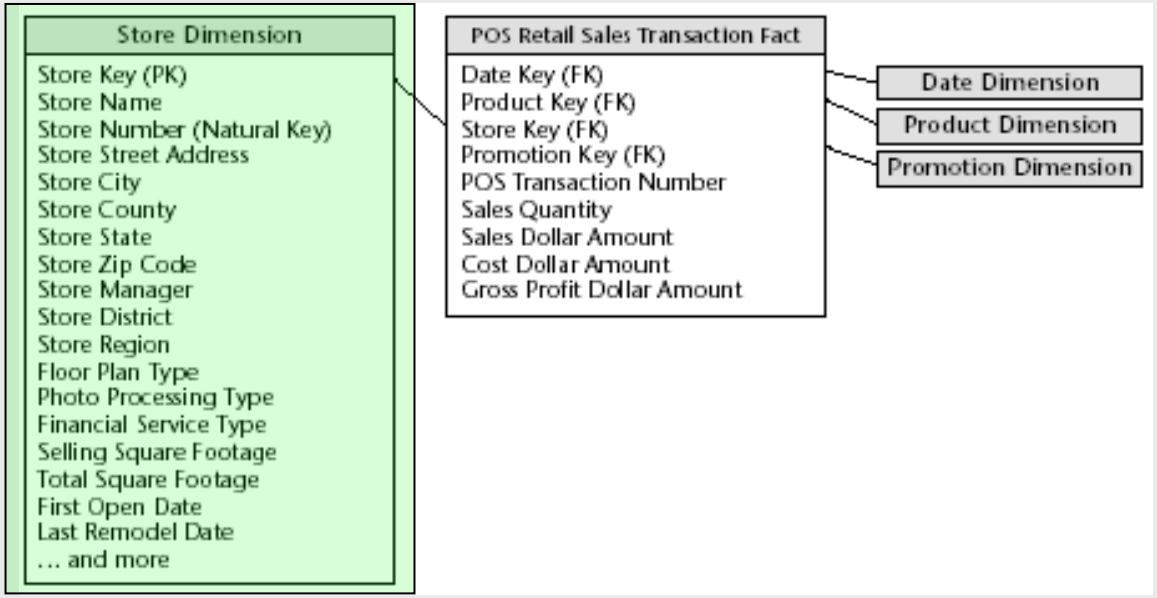
# Dimension Table Attributes

- **Product Dimension**
  - **Example content of a Product dimension table**

| Product Key | Product Description | Brand Description | Category Description | Department Description | Fat Content |
|---|---|---|---|---|---|
| 1 | Baked Well Light Sourdough Fresh Bread | Baked Well | Bread | Bakery | Reduced Fat |
| 2 | Fluffy Sliced Whole Wheat | Fluffy | Bread | Bakery | Regular Fat |
| 3 | Fluffy Light Sliced Whole Wheat | Fluffy | Bread | Bakery | Reduced Fat |
| 4 | Fat Free Mini Cinnamon Rolls | Light | Sweeten Bread | Bakery | Non-Fat |
| 5 | Diet Lovers Vanilla 2 Gallon | Coldpack | Frozen Desserts | Frozen Foods | Non-Fat |
| 6 | Light and Creamy Butter Pecan 1 Pint | Freshlike | Frozen Desserts | Frozen Foods | Reduced Fat |
| 7 | Chocolate Lovers 1/2 Gallon | Frigid | Frozen Desserts | Frozen Foods | Regular Fat |
| 8 | Strawberry Ice Creamy 1 Pint | Icy | Frozen Desserts | Frozen Foods | Regular Fat |
| 9 | Icy Ice Cream Sandwiches | Icy | Frozen Desserts | Frozen Foods | Regular Fat |

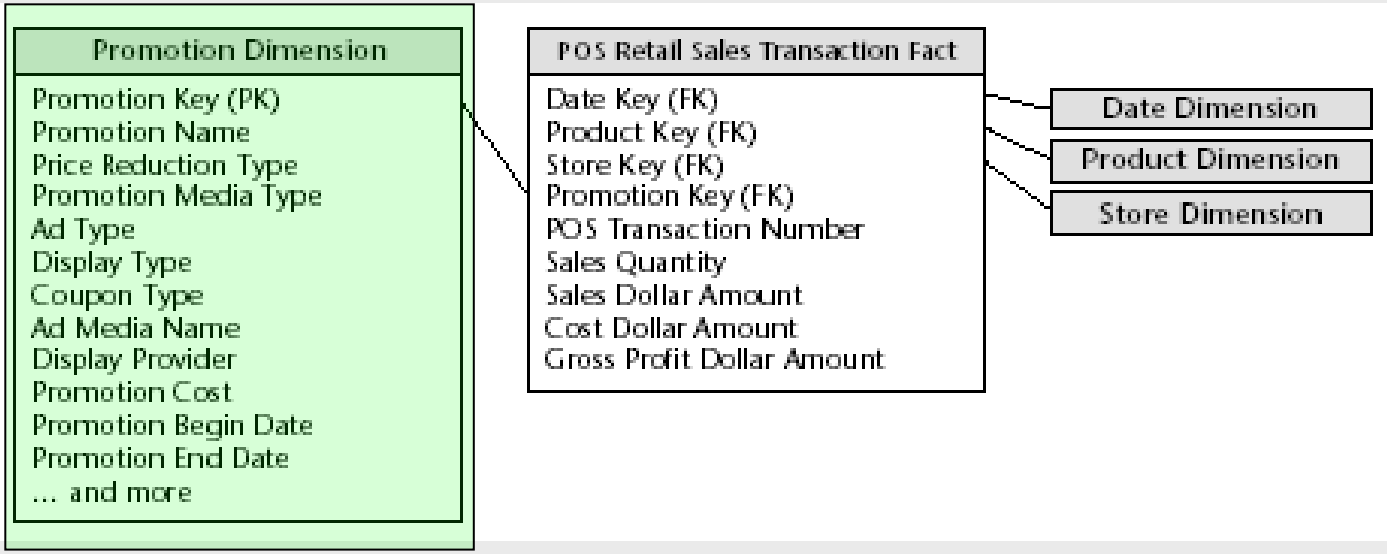
# Dimension Table Attributes

- **Store Dimension**
  - **Describes every store in our grocery chain**
  - **Is the primary geographic dimension in our case study**
    - Each store can be thought of as a location
    - We can roll stores up to any geographic attribute, such as ZIP code, county, and state in the United States
  - **Store dimension in the retail sales schema**



# Dimension Table Attributes

- **Promotion Dimension**
  - **Is potentially the most interesting dimension in our schema**
  - **Describes the promotion conditions under which a product was sold**
    - Temporary price reductions, end-aisle displays, newspaper ads, and coupons
  - **Promotion dimension in the retail sales schema**



# Retail Case Study: Complete Star-Schema

